

PetPulse

GitHub · <https://github.com/laila2005/PetPulse>

Computer Science · Faculty of Engineering & Computer Science



Semester: Spring 2025

PROJECT OVERVIEW

PetPulse is a comprehensive, AI-enhanced digital ecosystem that connects pet owners with veterinarians, trainers, and pet shops. It solves the fragmented pet care industry by centralizing bookings, marketplace, community, and health tracking into one secure and intuitive platform. Built as a full-stack graduation project at SUT.

10

Team Size

10 Weeks

Duration

React · Node.js

Tech Stack

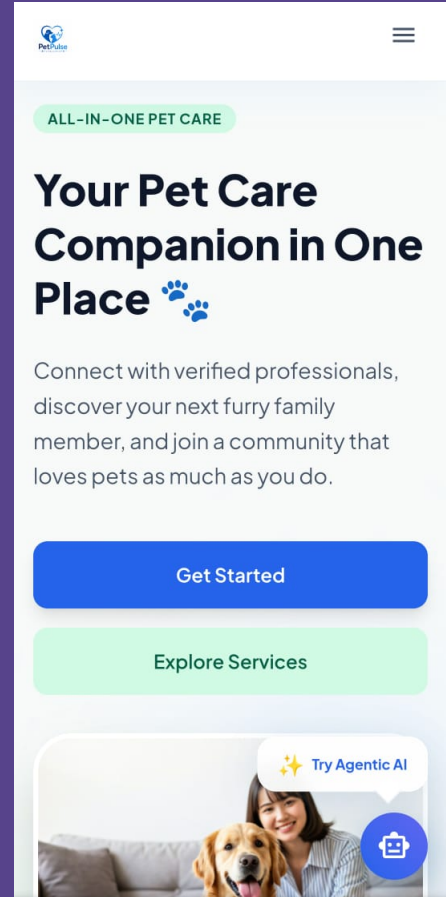


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TEAM MEMBERS

1

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Backend Developer

Computer Science

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Backend Production

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3

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Agentic AI

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7

Gaber Hosny

ID: 230101439

Scurity

Cyber Security

8

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ID: 230101269

Agentic AI

Cyber Security

9

Mahmoud Mustafa Mahmoud

ID: 230105193

Security

IT

10

Abdelrahman Essam Ali

ID: 230102357

Cyber Security

Cyber Security

02 PROBLEM STATEMENT

THE PROBLEM

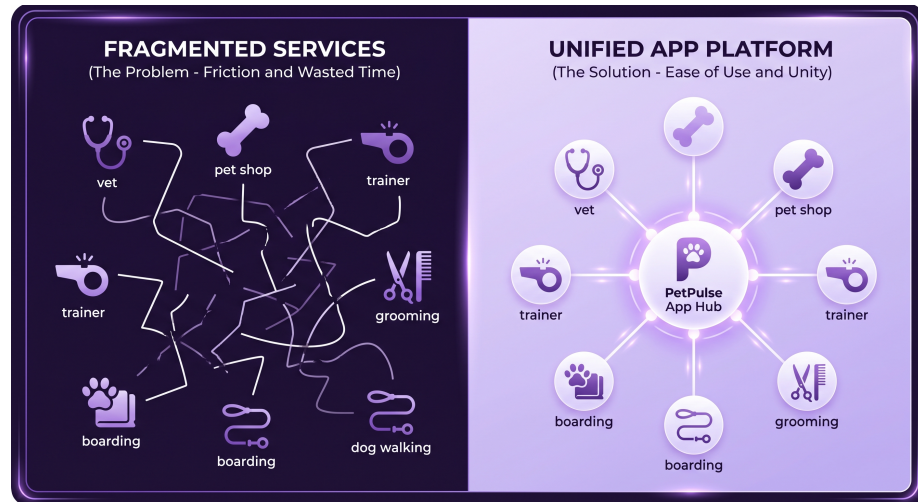
Pet ownership is fragmented across dozens of disconnected apps and services. Finding trusted vets, booking trainers, buying supplies, and connecting with pet communities all require separate platforms — creating friction, wasted time, and missed care opportunities.

WHY IT MATTERS

Impact 1: No single platform integrates veterinary booking, pet shops, training, and community features.

Impact 2: Millions of pet owners waste hours searching for reliable local pet care providers.

Impact 3: Lost & Found cases have low recovery rates without a geo-alert system.



70%

API Endpoints

50+

User Roles Supported

3+

User Roles

03 PROPOSED SOLUTION

PetPulse is an AI-enhanced full-stack web platform that unifies all pet care services — connecting pet owners with vets, trainers, pet shops, and a community — into one secure, intuitive ecosystem.



Geocentered Booking System

Pet owners book local veterinarians and trainers via integrated Leaflet maps with nearest-first ordering using Haversine distance calculations. Complete scheduling pipelines with admin approval workflows.



E-Commerce Marketplace

Verified pet shops list products on a full marketplace with browsing, filtering, cart, and Paymob payment integration. Vendor profiles require admin approval to ensure quality and trust.



VetAI Chatbot + Community

An adaptive AI chatbot uses function-calling to automate bookings on mobile. A real-time community feed with reactions, Lost & Found geo-alerts, and adoption boards complete the social layer.

1

React SPA Frontend

Vite-powered React 18 SPA with TailwindCSS. Handles auth, booking UI, marketplace, community feed, and VetAI chatbot.

2

Express REST API

Node.js + Express.js backend with 50+ endpoints. JWT authentication, input validation middleware, and RBAC for 5 user roles.

3

PostgreSQL Database

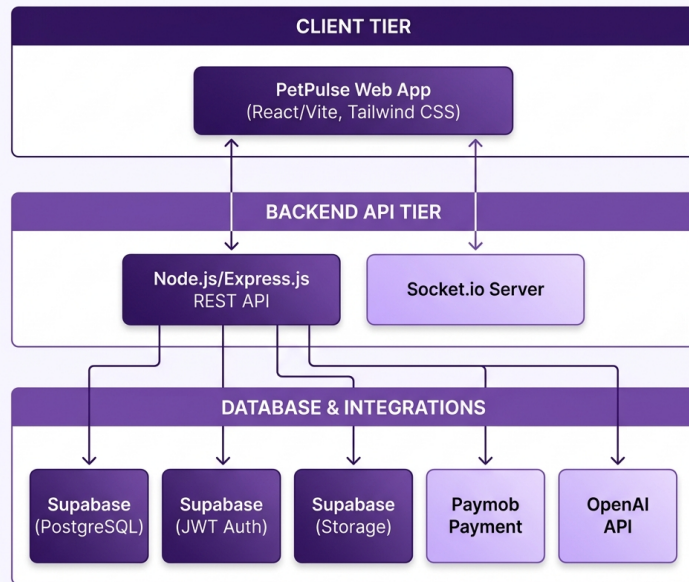
Relational DB with custom connection pooling (max:5) for Supabase. Parameterized queries prevent SQL injection. Complex joins for bookings, reviews, products.

4

External Integrations

Cloudinary for secure media uploads. Leaflet Maps for geo-bookings. Paymob for payments. Vercel for deployment.

PETPULSE PLATFORM SYSTEM ARCHITECTURE



TOOLS & TECHNOLOGIES

• Express.js • PostgreSQL • JWT • bcrypt • Cloudinary • Leaflet • Paymob • Vercel

VALUE PROPOSITIONS

- We provide an all-in-one platform that connects pet owners with everything their pets need.
- This includes access to pet food (dry food), pet supplies, veterinary services, and clinics.
- The platform simplifies communication between pet owners and veterinarians, making it easier to book appointments and get consultations.
- We also ensure fast and reliable access to emergency veterinary services, helping pet owners find immediate care in urgent situations.
- In addition, we aim to build a strong community where pet owners can connect, share experiences, and support each other.

COST STRUCTURE

The platform will incur costs in the following:

- website development
- Hosting and servers
- Marketing and advertising

KEY ACTIVITIES

- Platform development and maintenance
- Managing bookings and payment processes
- Partner onboarding (veterinarians, clinics, and brands)
- Marketing and user acquisition

KEY PARTNERS

The success of the platform depends on strong partnerships with:

- Veterinary clinics and hospitals
- Pet product suppliers
- Payment gateways such as Paymob

TARGET CUSTOMERS

Our platform is designed for:

- Pet owners
- Veterinarians
- Veterinary clinics
- Veterinary hospital

MARKETING & CUSTOMER ACQUISITION

We will attract users through:

- Social media advertising (Instagram, TikTok)
- Influencer marketing in the pet niche
- Partnerships with veterinary clinics

REVENUE MODEL

- We generate revenue through multiple streams:
- Commission-Based Model:
 - We take a percentage from each transaction made through the platform (clinic bookings, services).
- Advertising:
 - Revenue from ads placed by veterinarians, clinics, or pet-related brands and products.
- Subscription Plans (Optional):
 - Premium monthly or yearly subscriptions

DESIGNED FOR :

DESIGNED BY :

DATE :

VERSION :

CUSTOMER RELATIONSHIP

We aim to build and maintain strong relationships with users through:

Web - based chat with veterinarians
Real-time chat and updates for instant communication
Notifications (appointments, vaccinations, offers)
AI-assisted support for quick and intelligent responses

COMPETITIVE ADVANTAGE

- Fast and reliable emergency veterinary services
- Emergency routing that connects users to the nearest available care instantly
- An all-in-one solution for pet care needs
- A combined ecosystem of services and community in one platform
- Generative AI booking assistant that simplifies scheduling and service selection
- AI-assisted support for fast, intelligent, and automated user assistance
- Advanced vet discovery features to help users find the best veterinarians and clinics
- Real-time chat and updates for instant communication between users and service providers
- Lost & Found system to help users report and locate missing pets
- Access to extended services such as pet adoption and grooming

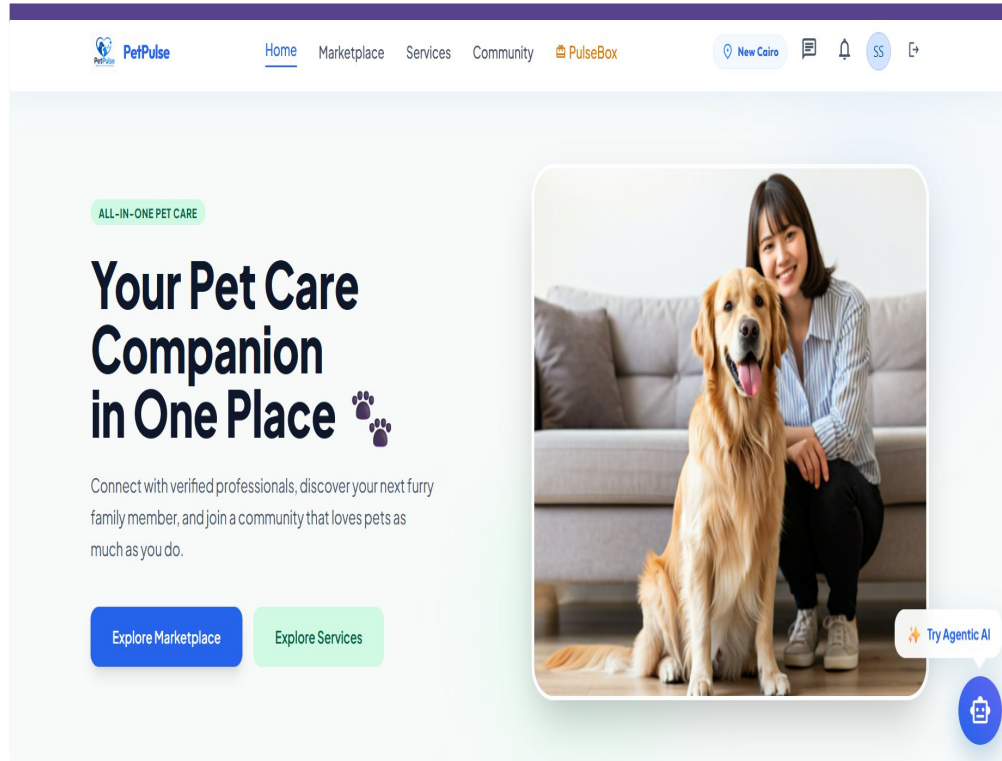
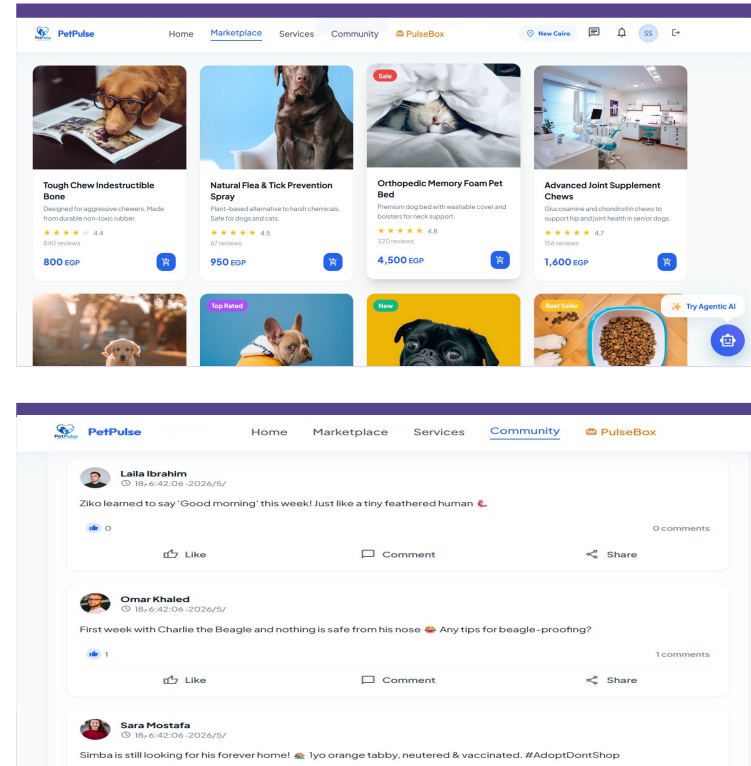


Fig. 1 — Homepage: Marketplace with verified vendors and product listings

Demo Link: <https://youtu.be/Z-4jVZxJzlo>

100%

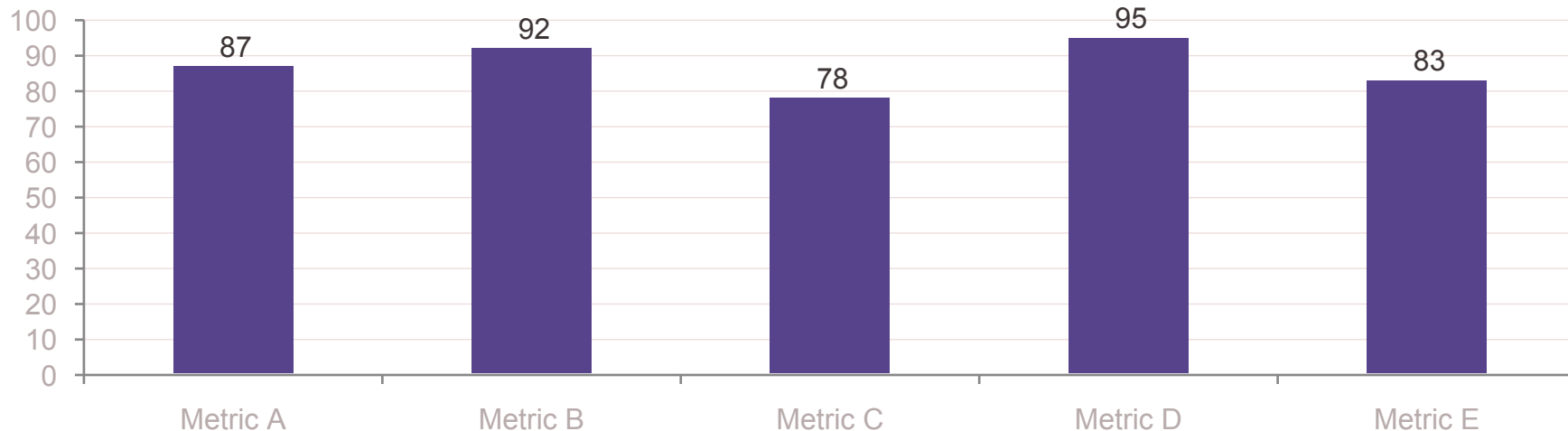
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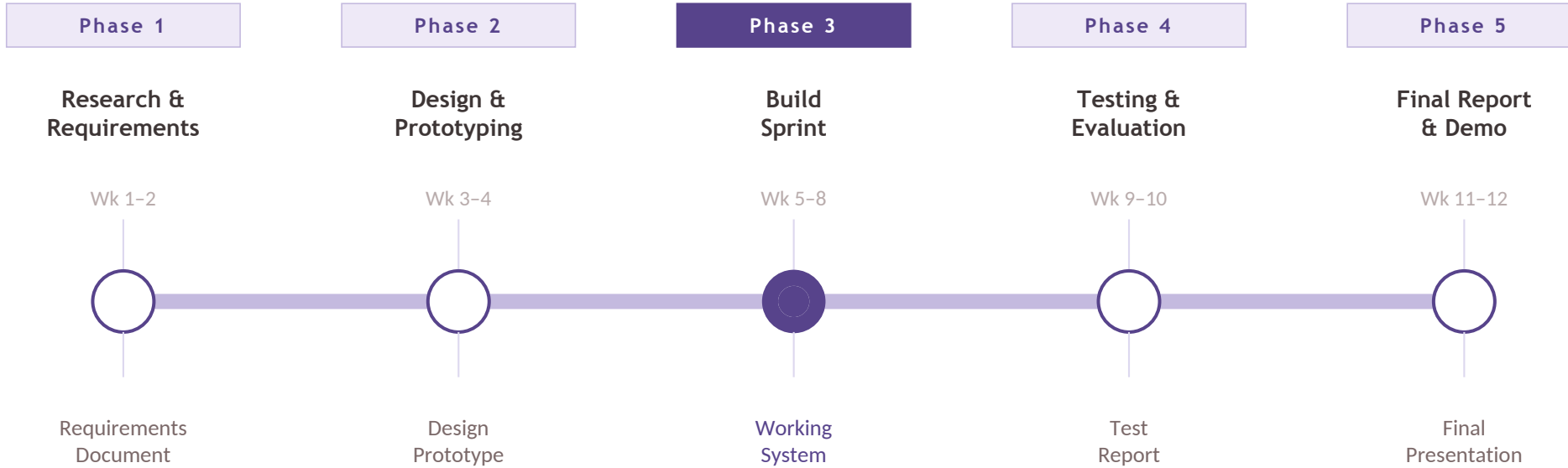
50+

REST API Endpoints

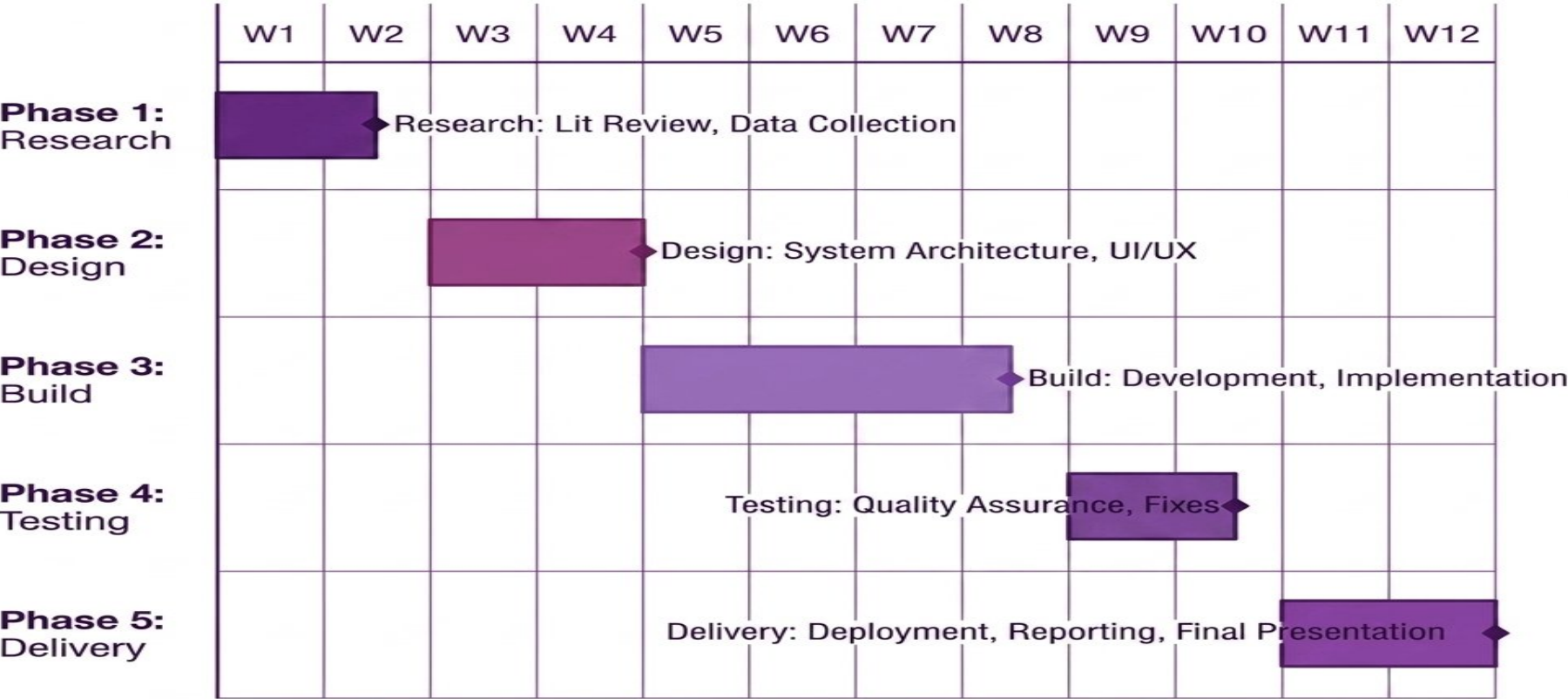
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User Roles (RBAC)





Timeline: Week 1 to Week 12



Thank You

We welcome your questions

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